

PROJECT REQUIREMENTS & SCOPE OF WORK

Battery Storage Economic Viability Calculator

Web Application for MySolar-PV / NGen Battery Systems



Prepared By	Faizan — Markhor Systems, Islamabad
Date	March 2025
Version	v1.0 — For Review and Approval
Project Type	Custom Web Application
Primary Market	Germany (DACH Region)

This document defines the full requirements, scope, and timeline for the Battery Storage Calculator project. It serves as a shared reference for both parties throughout development.

SECTION 01

| Project Overview

This project delivers a professional, web-based Battery Storage Economic Viability Calculator, hosted on a dedicated web space and integrated into the MySolar-PV website. The tool models financial returns for battery storage systems ranging from residential (10 kWh) to commercial (100 kW+) scale, across all relevant German energy market use cases.



100 Megawatt

The tool is built around two strategic goals:

- ▶ Educate — Show customers the real financial potential of battery storage, with and without grid balancing participation, using their own numbers.
- ▶ Convert — Allow interested users to download a branded PDF report and optionally leave contact details for a follow-up offer. No pressure, no forced registration.



if or if not, we decide this later

SECTION 02

Objectives



see extra text 1 on page 9

- ▶ Model revenue across all 6 use cases: self-consumption, arbitrage, peak shaving, PRL, SRL, and VPP participation.
- ▶ Demonstrate that balancing market participation (PRL/SRL/VPP) delivers approximately 40% higher returns than standard operation.
- ▶ Allow interactive scenario testing — users switch battery size and see updated results instantly.
- ▶ Generate a professional, print-ready DIN A4 PDF report with MySolar-PV branding and a revenue pie chart.
- ▶ Capture optional leads (name, email, phone) without pressure or mandatory registration.
- ▶ Build architecture that supports future addition of the multi-tenant installer white-label system.



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SECTION 03

Target Users



see extra text 2 on page 10

Residential Customers

Homeowners considering small battery systems (10–84 kWh) combined with PV panels. Interested in electricity bill savings and potential VPP income of €400–500/month from grid stabilisation services.

Commercial Customers



see extra text 3 on page 11

Businesses with significant electricity demand looking for peak shaving, load optimisation, and reserve energy services. Battery range: 100 kW and above.

Installer Partners (Phase 2)

Certified installation companies supplied by MySolar-PV who want to offer the calculator to their own end customers under their own branding and domain.

SECTION 04

Full Feature List

FEATURE	DESCRIPTION	PHASE
Landing Page	Branded entry point with clear CTA	Phase 1
Multi-step Questionnaire	Tick applicable use cases — controls which modules appear	Phase 1
Technical Input Form	Battery kW/kWh, energy source, PV/wind specs	Phase 1
Financial Input Form	Electricity price, annual bill, system purchase cost	Phase 1
Region Selector	North / Centre / South Germany solar yield adjustment	Phase 1
Load Profile Upload	CSV upload — detects consumption peaks automatically	Phase 1
Manual Load Fallback	Usage pattern selector: morning / noon / evening / night	Phase 1
Calculation Engine	6-stream revenue model using historical market averages	Phase 1
Scenario Comparison	Instantly switch battery size and see updated results	Phase 1
Results Dashboard	Charts, 15-year projection, key financial KPIs	Phase 1
Revenue Pie Chart	Visual income split by use case	Phase 1
Balancing Market Callout	Prominent display of ~40% uplift from PRL/SRL/VPP	Phase 1
PDF Report Generator	Branded DIN A4 printable report with embedded charts	Phase 1
Soft Lead Capture Form	Optional name / email / phone — zero mandatory fields	Phase 1
Lead Email Notification	Auto-email to Stephan on each lead submission	Phase 1
Wind Turbine Mode	No local consumption — storage modelled at turbine base	Phase 1
Multi-tenant Admin Panel	Create and manage installer partner accounts	Phase 2
Installer White-Label	Custom logo and brand colours per installer	Phase 2
Access Toggle	Activate / deactivate any partner's tool access instantly	Phase 2



See Text 5 Page 13

SECTION 05

Calculation Engine

The calculator uses pre-loaded historical market averages baked into the application at build time. No live API integrations are required for Phase 1, keeping the system fast, reliable, and cost-free to operate. Data can be updated periodically as market conditions change.

All six revenue streams are calculated simultaneously. Users can enable or disable individual streams via the questionnaire to see the exact impact each one has on total returns.

PRL Revenue	Historical average prices from regelleistung.net (€/MW/week) applied to available battery capacity based on selected use cases.
SRL / aFRR Revenue	Historical average contracted rates from regelleistung.net applied to pre-qualified capacity with realistic availability assumptions.
EPEX Arbitrage	Historical day-ahead price spread modelled across realistic daily charge/discharge cycles with round-trip efficiency losses factored in.
Self-Consumption	User-entered electricity price × kWh offset by battery, adjusted for 90% round-trip efficiency and seasonal variation.
Peak Shaving	Demand charge reduction estimated from uploaded load profile peaks or manual consumption pattern input.
VPP Participation	Fixed programme rate based on NGen/MySolar-PV VPP payouts (~€400–500/month per 41.9 kWh unit enrolled in the programme).
Solar Yield	Regional constants: 900 kWh/kWp/year (North Germany), 950 (Central), 1,000 (South Germany).



PLS THINK on Load - shifting

SECTION 06

User Flow

if or if not, we decide this later

1. Landing Page — User reads brief introduction and clicks 'Start Calculation'.
2. Questionnaire — User selects applicable use cases via checkboxes (controls visible modules).
3. Technical Inputs — Battery specs, energy source type, PV or wind details, region selection.
4. Financial Inputs — Current electricity price, annual bill, optional system purchase cost.
5. Calculation — Engine processes all inputs and renders the results dashboard.
6. Scenario Testing — User adjusts battery size with a slider and instantly sees updated numbers.
7. PDF Download — User downloads a branded, print-ready DIN A4 report.
8. Lead Form — Optional: user submits name, email, and/or phone to request a personalised offer.



See Text 4 on Page 12

SECTION 07

Technical Architecture

Frontend Framework	Next.js (React) — fast load times, SEO-ready, server-side rendering
Styling	Tailwind CSS — responsive design in MySolar-PV brand colours and typography
Calculation Engine	TypeScript — pure logic module, fully unit-testable, zero external dependencies
Charts	Recharts — interactive pie chart, bar chart, and 15-year projection line chart
CSV Parsing	PapaParse — client-side load profile upload and automatic peak detection
PDF Generation	React-PDF or Puppeteer — branded DIN A4 output with embedded charts and logo
Lead Backend	Node.js lightweight API — stores lead records, triggers email to Stephan on submission
Email Delivery	Resend or Nodemailer — automated notification with user inputs attached
Hosting	Dedicated VPS or Vercel — separate webspace as requested, custom domain ready
Phase 2 — Database	PostgreSQL — multi-tenant account management, installer records, access flags
Phase 2 — Auth	JWT-based authentication for the installer admin panel

SECTION 08

Responsibilities

8.1 Markhor Systems Delivers

- ▶ Complete Phase 1 web application as described in Section 4.
- ▶ Calculation engine calibrated with historical PRL/SRL/EPEX averages.
- ▶ Responsive design matching MySolar-PV brand colours and visual style.
- ▶ Branded PDF report template with placeholder for client logo and colours.
- ▶ Lead capture backend and automated email notification system.
- ▶ Deployment to dedicated webspace / hosting environment.
- ▶ Post-delivery handover documentation and integration guide for Stephan's web specialist.
- ▶ 30 days post-delivery bug fix support at no additional charge.

8.2 MySolar-PV / Stephan Provides

- ▶ Company logo in high resolution (PNG or SVG format).
- ▶ Brand colour codes (HEX values) and any font preferences.
- ▶ NGen product specifications: available kWh/kW configurations and efficiency ratings.
- ▶ VPP programme payout rates to calibrate the VPP revenue stream.
- ▶ Historical PRL/SRL rate data if available (otherwise Markhor Systems sources from regelleistung.net).
- ▶ Access credentials for hosting / webspace, or approval for Markhor Systems to provision hosting.
- ▶ Timely milestone review and feedback within 3 business days to maintain project timeline.

SECTION 09

Milestones & Timeline

Total delivery timeline: 5 to 6 weeks from project kickoff. Timeline assumes timely feedback at each milestone (within 3 business days). Delays in asset delivery or approvals will extend the schedule accordingly.

#	MILESTONE	DELIVERABLE	TIMELINE
1	Project Kickoff	Requirements confirmed, brand assets received	Week 1
2	UI/UX Prototype	Clickable wireframe submitted for client approval	Week 1–2
3	Input Forms & Questionnaire	All input screens functional and styled	Week 2–3
4	Calculation Engine (Core)	All 6 revenue streams calculating correctly	Week 3
5	Results Dashboard & Charts	Interactive results with scenario switcher live	Week 4
6	PDF Report Generator	Branded DIN A4 report downloadable	Week 4
7	Lead Capture & Email System	Lead form live, email notifications operational	Week 5
8	QA, Testing & Deployment	Live on dedicated webspace, cross-browser tested	Week 5–6

SECTION 10

Commercial Terms

DELIVERABLE	TIMELINE	INVESTMENT
Phase 1 — Full Calculator Web Application (all features in Section 4)	5–6 Weeks	€1,250
Phase 2 — Multi-Tenant Installer White-Label System (optional add-on)	+2–3 Months	€700
Monthly Maintenance Retainer — optional ongoing support	Monthly	€150
TOTAL (Phase 1 Only)	5–6 Weeks	€1,250

10.1 What Is Included

- ▶ All development work as described in Sections 4 through 7.
- ▶ Deployment to webspace and initial server configuration.
- ▶ Integration documentation for Stephan's web specialist.
- ▶ 30 days post-delivery bug fix support at no additional charge.

10.2 What Is Not Included

- ▶ Ongoing hosting or domain costs (if Markhor Systems provisions hosting, billed at cost).
- ▶ Live market data API subscriptions (not required for Phase 1).
- ▶ Translation into other languages.
- ▶ Feature changes after a milestone has been approved (change requests quoted separately).

SECTION 11

Future Roadmap

The following items are explicitly outside the scope of Phase 1 but are planned for future development. The Phase 1 architecture is designed to accommodate these additions without requiring a full rebuild.

- ▶ Multi-tenant installer portal with white-label branding per partner account (Phase 2).
- ▶ Voluntary contribution / donation feature for the public-facing version of the tool.
- ▶ 'Which storage is right for me?' recommendation engine based on questionnaire answers.
- ▶ Live market data integration (EPEX Spot, regelleistung.net API) for real-time revenue estimates.
- ▶ Multi-language support: English and German.
- ▶ NGen product recommendation engine — suggests optimal unit configuration based on user inputs.

Developer	Markhor Systems — Faizan
Document Date	March 2025
Document Version	v1.0

Extra Text 1:

Revised Draft: Use Case – Load Shifting

"We are addressing a total of seven use cases, with **load shifting** being a key addition. This involves a voluntary decision by the operator to bypass the fixed feed-in tariff (e.g., Renewable Energy Sources Act) and instead buffer the energy in storage. This stored energy is then fed into the grid at a later time—such as at night—to capitalize on potentially higher remuneration offered by dynamic, flexible tariffs.

To accurately assess the profitability of this shift, any applicable grid fees must be taken into account. Consequently, we require three specific data fields to model this scenario:

1. **Dynamic Feed-in Tariff:** The potential remuneration rate for nighttime or flexible-period injection.
2. **Grid Fees:** The costs incurred per kWh for grid usage, which must be deducted from the gross revenue.
3. **Standard Feed-in Tariff:** The baseline remuneration the operator would regularly receive from the grid provider or via existing legislation (e.g., EEG).

Extra Text 2:

Technical Concept: Investment Return Visualization for Homeowners

"After completing the calculation, homeowners should be presented with a clear visual breakdown—ideally a **pie chart**—illustrating the different components of their return on investment (ROI) for a battery storage system.

1. Self-Consumption Optimization

One primary driver of returns is the optimization of self-generated energy. To calculate this, the purchase price of grid electricity must be specified.

2. Temporal Load Shifting (Energy Arbitrage)

A second revenue stream—**Load Shifting**—must be clearly visualized. This applies when customers receive no regular remuneration during midday peak hours (approx. 10:30 AM to 2:30 PM). The system calculates how much energy can be transferred to the second half of the day to be sold more profitably in the evening.

- *Note: Private households do not typically generate additional income through peak shaving.*

3. Energy Trading & Grid Services (Capacity Requirements)

Additional capacity for **spot market trading** or **balancing energy** (Regelenergie) is only viable if the battery is oversized by at least 50% of the actual requirement. Participation in these services must not jeopardize the primary goal of self-consumption optimization.

- **System Prompt:** During data entry, the user should see a disclaimer in parentheses: *"Note: This revenue stream requires a generously dimensioned battery capacity."*
- **Visualization:** The pie chart should explicitly show the potential share of income generated from balancing energy to give the customer a realistic overview.

4. VPP Integration (Virtual Power Plant)

The option to join a **Virtual Power Plant (VPP)** should be presented via a simple checkbox:

"Would you like to participate in the VPP?"

If the user selects "Yes," the following logic is applied in the background: By joining the community, the user's trading volumes across all sectors increase. Due to better collective terms, the user earns approximately **12% more** in both the spot market and balancing energy sectors. The system automatically switches between different markets via the VPP to maximize the overall effect.

The Goal: The primary objective is not just to show higher earnings, but to educate the customer: *"This company provides significant added value simply through the availability of a VPP."*

Key Terminology Used:

- **Self-Consumption Optimization:** Optimierung der eigenen Energie.
- **Load Shifting / Energy Arbitrage:** Lastverschiebung / zeitversetzter Verkauf.
- **Balancing Energy:** Regelenergie.
- **Spot Market Trading:** Börsenhandel.
- **Oversized / Generously dimensioned:** Üppig bemessen.

Extra Text 3:

Technical Concept: Industrial Peak Shaving Integration

"At the end of the day, these industrial clients should also be presented with a **pie chart**, with the key difference being the addition of an extra revenue stream: **Peak Shaving**.

To calculate this, customers must provide several key data points from their industrial energy supply contract.

1. Energy Consumption Input

We require the customer to specify their annual energy consumption. This section must include a dropdown menu allowing commercial users to select their unit of measurement:

- Kilowatt-hours (kWh)
- Megawatt-hours (MWh)
- Gigawatt-hours (GWh)

Example: If a customer enters 120 million kWh of annual energy consumption, the system should correctly process this as 120,000 MWh or 120 GWh.

2. Cost and Potential Savings

Further required inputs include:

- The **price per kW** (Demand Charge).
- The **grid fees** per kWh (Network Charges).

We then present the following question to the user:

"By what percentage will your grid fees be reduced if you smooth your demand profile and filter out consumption peaks?"

Typically, a customer might indicate a potential saving of **75% of their grid fees** through effective load smoothing. Based on this load profile, the system will calculate the required battery storage size.

3. Sizing and Safety Margins

While these figures are non-binding, we must provide a recommended order of magnitude. This recommendation should be calculated so that the storage system does not reach its technical limits while performing peak shaving. On the contrary, the suggested battery capacity should be **at least 50% larger** than the actual technical requirement.

The financial stakes here are significant: a customer with this level of consumption can potentially save approximately **€1 million per year** by implementing high-quality peak shaving strategies.

4. Mandatory Disclaimer

At this stage, we must display a clear warning:

"Notice: The final calculation and recommendation for the storage size must be conducted by a qualified engineering firm. Please contact our specialized department for a detailed consultation."

Key Terminology Note:

- **Peak Shaving:** Eliminating high spikes in energy demand.
- **Demand Charge / Price per kW:** *Leistungspreis* (what industrial customers pay for their peak load).
- **Grid Fees:** *Netzentgelte*.
- **Load Smoothing:** *Abnahme glätten*.

Text 4 :

Translation: Segmented Lead Calculator Concept

I want to start a new text, which should be translated into English as follows:

I want to design it so that the person using the calculator is asked a question right at the beginning: **Are you a private user? Or are you an installer? Are you a company looking to run an optimization calculation for yourself?**

We collect this data right at the start and keep it "in the back of our minds" for the duration of the session.

If the customer enters as an **installer**, they will be asked the following questions while running through the calculator:

- Would you like to offer this software to your customers under your own label?
- Everything branded to your company?
- Even the contact page would feature your phone number and email address!
- Would you like to use this page and this calculator for your own advertising?

Apply now as a marketer and installer and enter the market with a fully professional calculation tool for your customers.

With this calculator, you won't just generate exclusive inquiries; you will have customers who want nothing else but this specific technology! **Create a competitive advantage for yourself!**

Get this calculator at favorable conditions to promote it as often and as much as you like across all imaginable media! Use this calculator as your ultra-cost-effective lead machine – **become a retail partner today and contact our specialist department.**

.-----.

If a dealer/customer applies here, we must be able to store an email address in the backend where these inquiries land and can then be processed by us.

The customer must actually state what they are at the very beginning. However, they do **not** have to provide a name, street, or address yet. They should first experience the great performance of the calculator.

- The **private individual** is not bothered with entering address data at this stage.
- A **company** looking to do an evaluation for themselves is also not bothered with address entry yet.

But the **installer** is clearly asked by us whether they are an installer. The question could be: **"Are you an installer and provider of photovoltaic systems?"**

If they answer "Yes," they will see exactly the same program as everyone else. The only difference is that, in between, an attempt is made to recruit them to become an installer for us and install our technology.

The installer will recognize a massive advantage in using this calculator as their "alternator" (lead generator). We will then actually provide the installer with a slightly slimmed-down version of this calculator so they can serve private individuals.

And perhaps we will set it up so that the installer pays a small monthly fee for their own calculator and lead generator anyway.

Text 5

„It must be possible to input additional energy sources, such as one or two wind turbines, as well as biomass power plants. If the user specifies which performance data is available at what time—whether entered manually or via a load profile—it should then be possible to simulate the input of this energy at different points in time.

There is one aspect I am not 100% certain about yet, but I want to address it: the possibility of transferring energy from biomass or wind power into a large buffer storage unit. This storage usually provides balancing energy (*Regelenergie*). It should be possible to feed a portion of this energy into the arbitrage and trading markets to sell it off. However, perhaps a slider could be used to reserve a certain amount of energy for balancing energy support. This could be an exceptionally profitable revenue stream if there is a local demand for delivering large amounts of energy at short notice during unusual times.

While I cannot yet confirm if this constitutes a viable revenue market, we should at least allow users to input energy sources like biogas or wind and 'unlock' their marketing options within the VPP (Virtual Power Plant) via NGen's own control center.

I believe that sooner or later, it will also be possible to simulate the sale of large energy volumes based on historical data. This would allow a farmer or wind farm operator to input their data to see what additional revenue could be generated by installing one of our storage systems alongside their turbines. This would make it possible to calculate and demonstrate the economic viability of relatively large storage units, ranging from 2 MW to 5 MW, 10 MW, or even 20 MW."

Key Terms Used:

- **Balancing Energy:** *Regelenergie* (crucial for grid stability).
- **Arbitrage Market:** Buying/selling based on price differences.
- **VPP (Virtual Power Plant):** *Virtuelles Kraftwerk*.
- **Economic Viability:** *Wirtschaftlichkeit*.
- **Control Center:** *Leitwarte*.
-